

A Local Structure Theory of the EABC Grid Modulo 12

Collaborative Draft

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Abstract

We introduce the modulo-12 EABC grid

$$Q_n = (12n - 11, 12n - 7, 12n - 5, 12n - 1), \quad n \in \mathbb{N},$$

and show that it provides a complete local description of small prime constellations above 3. In this framework, all prime twins > 3 lie in exactly two linear channels, all dense prime triplets are forced by the singular anchor 3, block-crossing prime quadruplets are precisely the double activation points of both twin channels, and dense prime quintuplets are exactly the oriented extensions of such double points.

1 The EABC grid

Definition 1 (EABC families). For $n \in \mathbb{N}$, define

$$E_n := 12n - 11, \quad A_n := 12n - 7, \quad B_n := 12n - 5, \quad C_n := 12n - 1.$$

We call

$$Q_n := (E_n, A_n, B_n, C_n)$$

the n -th *EABC block*.

Lemma 2 (Prime decomposition modulo 12). *Every prime $p > 3$ lies in exactly one of the four families*

$$E, \quad A, \quad B, \quad C.$$

Proof. A prime $p > 3$ is divisible by neither 2 nor 3. Hence

$$p \equiv 1, 5, 7, \text{ or } 11 \pmod{12}.$$

These four residue classes are parametrized disjointly and exhaustively by

$$12n - 11, \quad 12n - 7, \quad 12n - 5, \quad 12n - 1.$$

□

Corollary 3. *The set of primes > 3 decomposes as*

$$\mathbb{P} \setminus \{2, 3\} = (\mathbb{P} \cap E) \dot{\cup} (\mathbb{P} \cap A) \dot{\cup} (\mathbb{P} \cap B) \dot{\cup} (\mathbb{P} \cap C).$$

2 The two twin channels

Definition 4 (Twin channels). For each EABC block Q_n , define the internal and external twin channels by

$$\begin{aligned} T_n^{\text{int}} &:= (A_n, B_n) = (12n - 7, 12n - 5), \\ T_n^{\text{ext}} &:= (C_n, E_{n+1}) = (12n - 1, 12n + 1). \end{aligned}$$

We further define the *twin signature*

$$\tau(Q_n) := (\mathbf{1}_{T_n^{\text{int}} \text{ is a twin prime pair}}, \mathbf{1}_{T_n^{\text{ext}} \text{ is a twin prime pair}}) \in \{0, 1\}^2.$$

Lemma 5 (Completeness of the twin channels). *Every twin prime pair $(p, p + 2)$ with $p > 3$ lies uniquely in exactly one of the two patterns*

$$(12n - 7, 12n - 5) \quad \text{or} \quad (12n - 1, 12n + 1).$$

Proof. For $p > 3$, a twin prime pair can only have residue pattern

$$(5, 7) \quad \text{or} \quad (11, 1) \pmod{12}.$$

The first case is exactly

$$(A_n, B_n),$$

and the second case is exactly

$$(C_n, E_{n+1}).$$

Uniqueness follows from the linear parametrization. □

Corollary 6 (Classification of twin primes > 3). *All twin primes > 3 are exactly the prime realizations of the two linear forms*

$$(12n - 7, 12n - 5) \quad \text{and} \quad (12n - 1, 12n + 1).$$

3 Dense prime triplets

Definition 7 (Dense prime triplets). A *dense prime triplet* is a prime triple of one of the two forms

$$(p, p + 2, p + 6) \quad \text{or} \quad (p, p + 4, p + 6).$$

Lemma 8 (Every dense triplet contains 3). *Every dense prime triplet necessarily contains the prime 3.*

Proof. In either pattern

$$p, p + 2, p + 6 \quad \text{or} \quad p, p + 4, p + 6,$$

at least one of the three numbers is divisible by 3. If all three are prime, that number must be 3. □

Corollary 9 (Classification of dense triplets). *The only dense prime triplets are*

$$(3, 5, 7) \quad \text{and} \quad (3, 7, 11).$$

Remark 10. Thus dense triplets are not free internal configurations of the EABC grid. They are singular boundary phenomena forced by the exceptional prime 3.

4 Double points and prime quadruplets

Definition 11 (Double point). An EABC block Q_n is called a *double point* if

$$\tau(Q_n) = (1, 1).$$

Equivalently, this means that

$$A_n, B_n, C_n, E_{n+1}$$

are all prime.

Lemma 12 (Double points are block-crossing quadruplets). *An EABC block Q_n is a double point if and only if*

$$(12n - 7, 12n - 5, 12n - 1, 12n + 1)$$

is a prime quadruplet.

Proof. The condition $\tau(Q_n) = (1, 1)$ means that both

$$(12n - 7, 12n - 5) \quad \text{and} \quad (12n - 1, 12n + 1)$$

are twin prime pairs. This is equivalent to the primality of all four numbers

$$12n - 7, 12n - 5, 12n - 1, 12n + 1,$$

which form the pattern

$$p, p + 2, p + 6, p + 8.$$

□

Corollary 13. *The block-crossing prime quadruplets are exactly the simultaneous activations of the two twin channels.*

5 Dense prime quintuplets

Definition 14 (Dense prime quintuplet). A *dense prime quintuplet* is a prime constellation of the form

$$(p, p + 2, p + 6, p + 8, p + 12).$$

Lemma 15 (Two EABC forms of quintuplets). *Every dense prime quintuplet with $p > 3$ has exactly one of the two forms*

$$(A_n, B_n, C_n, E_{n+1}, A_{n+1})$$

or

$$(C_n, E_{n+1}, A_{n+1}, B_{n+1}, C_{n+1})$$

for a unique $n \in \mathbb{N}$.

Proof. For $p > 3$, only the starting residues 5 and 11 modulo 12 are possible.

If

$$p \equiv 5 \pmod{12},$$

then $p = A_n$, hence

$$p + 2 = B_n, \quad p + 6 = C_n, \quad p + 8 = E_{n+1}, \quad p + 12 = A_{n+1}.$$

If

$$p \equiv 11 \pmod{12},$$

then $p = C_n$, hence

$$p + 2 = E_{n+1}, \quad p + 6 = A_{n+1}, \quad p + 8 = B_{n+1}, \quad p + 12 = C_{n+1}.$$

The starting residues 1 and 7 are impossible, since then one of the subsequent terms would be divisible by 3. □

Lemma 16 (Every dense quintuplet contains exactly one double point). *Every dense prime quintuplet contains exactly one block-crossing prime quadruplet.*

Proof. In the first form

$$(A_n, B_n, C_n, E_{n+1}, A_{n+1}),$$

the first four terms form the quadruplet.

In the second form

$$(C_n, E_{n+1}, A_{n+1}, B_{n+1}, C_{n+1}),$$

the first four terms form the quadruplet.

A dense quintuplet has only five entries, so it cannot contain two distinct quadruplets of this type. $\square$

Corollary 17. *Every dense prime quintuplet is an oriented extension of a unique double point.*

6 Main theorem

Theorem 18 (Local completeness of the EABC grid). *The EABC grid*

$$Q_n = (12n - 11, 12n - 7, 12n - 5, 12n - 1)$$

provides a complete local description of small prime constellations above 3:

- (i) *Every prime $p > 3$ lies uniquely in one of the four families E, A, B, C .*
- (ii) *Every twin prime pair > 3 lies uniquely in one of the two channels*

$$(A_n, B_n) \quad \text{or} \quad (C_n, E_{n+1}).$$

- (iii) *Every dense prime triplet is one of the two singular 3-anchored cases*

$$(3, 5, 7), \quad (3, 7, 11).$$

- (iv) *Every double point $\tau(Q_n) = (1, 1)$ is exactly a block-crossing prime quadruplet*

$$(12n - 7, 12n - 5, 12n - 1, 12n + 1).$$

- (v) *Every dense prime quintuplet is exactly an oriented extension of such a double point, hence has one of the two forms*

$$(A_n, B_n, C_n, E_{n+1}, A_{n+1}) \quad \text{or} \quad (C_n, E_{n+1}, A_{n+1}, B_{n+1}, C_{n+1}).$$

Therefore the full local twin-, triplet-, quadruplet-, and quintuplet-structure of the prime grid modulo 12 is completely described.

Proof. Statements (i) and (ii) follow from the first two lemmas. Statement (iii) is the triplet corollary. Statement (iv) is the quadruplet lemma. Statement (v) follows from the two quintuplet lemmas. $\square$

Remark 19. In this picture, the prime quadruplet is the simultaneous activation of both twin channels, while the prime quintuplet is the first genuinely directed continuation of such a double point.